

TE Curve Verification Pipeline Directional Model Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report is the canonical TE Curve Verification Pipeline offline comparison between RCIM Model-Bank Reproduction , recovered original, retuned paper-reference model banks, and repository-owned Wave 1 and Wave 2.1 model candidates. It starts from the current direction-aware comparison matrix.

Dataset And Split

- dataset config: config/datasets/transmission_error_dataset.yaml ;
- dataset root: data\polished_dataset ;
- comparison mode: reduced_selected_model_matrix ;
- candidate count: 9 ;
- held-out curve count before candidate filtering: 100 ;
- percentage-error denominator: peak_to_peak_truth ;
- Fw candidates are evaluated only on forward curves;
- Bw candidates are evaluated only on backward curves;

Candidate Inventory

Candidate	Source	Family	Surface	Valid Directions
polished_rcim_model_bank_reproduction_GBM19_Fw	polished_rcim_model_bank_reproduction	GBM	Fw	forward
polished_rcim_model_bank_reproduction_ET19_Fw	polished_rcim_model_bank_reproduction	ET	Fw	forward
polished_feedforward_Fw	polished_model_development_registry	feedforward	Fw	forward
polished_tree_Fw	polished_model_development_registry	tree	Fw	forward
polished_harmonic_regression_Fw	polished_model_development_registry	harmonic_regression	Fw	forward
polished_periodic_gru_sequence_Fw	polished_model_development_registry	periodic_gru_sequence	Fw	forward
polished_periodic_mlp_harmonic_Fw	polished_model_development_registry	periodic_mlp_harmonic	Fw	forward
polished_wave4_3_mixture_density_k3_Fw	polished_model_development_registry	wave4_3_mixture_density_k3	Fw	forward
wave52b_offset_centered_shape_harmonic_Fw	wave52b_offset_harmonic_guided_registry	wave52b_offset_harmonic_guided	Fw	forward

Forward Comparison

Wave 5.2B Offset And Harmonic Guided Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_Fw	0.001695	0.002045	3.391	8.270

Polished Model-Development Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_periodic_gru_sequence_Fw	0.001195	0.001461	2.559	5.193
polished_wave4_3_mixture_density_k3_Fw	0.001528	0.001867	3.161	7.018
polished_periodic_mlp_harmonic_Fw	0.001735	0.002062	3.511	7.554
polished_tree_Fw	0.002125	0.002612	4.355	8.534
polished_feedforward_Fw	0.002130	0.002586	4.378	8.431
polished_harmonic_regression_Fw	0.003766	0.004246	8.148	15.111

Polished RCIM Model-Bank Reproduction Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_rcim_model_bank_reproduction_ET19_Fw	0.001155	0.001394	2.401	4.928
polished_rcim_model_bank_reproduction_GBM19_Fw	0.001644	0.001903	3.449	9.961

Curve Evidence

Each selected candidate is shown with the same four deterministic held-out operating conditions for this direction. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

Shared operating conditions

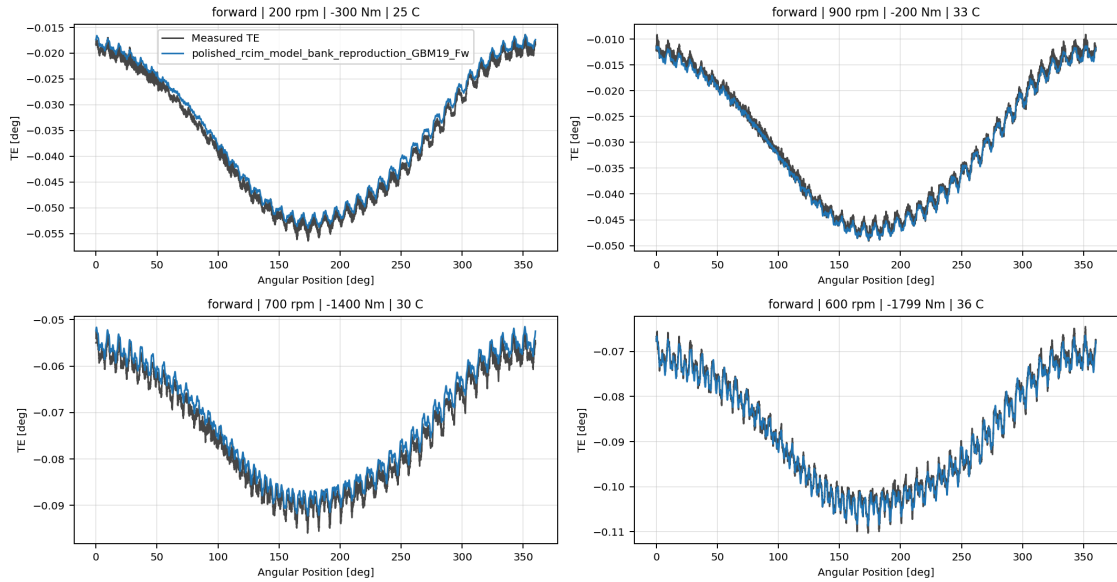
- 200 rpm, 300 Nm, 25 C
- 900 rpm, 200 Nm, 30 C
- 700 rpm, 1400 Nm, 30 C
- 600 rpm, 1800 Nm, 35 C

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
polished_rcim_model_bank_reproduction_GBM19_Fw	Selected candidate	0.001644	0.001903	3.449	9.961
polished_rcim_model_bank_reproduction_ET19_Fw	Surface leader	0.001155	0.001394	2.401	4.928
polished_feedforward_Fw	Selected candidate	0.002130	0.002586	4.378	8.431
polished_tree_Fw	Selected candidate	0.002125	0.002612	4.355	8.534
polished_harmonic_regression_Fw	Selected candidate	0.003766	0.004246	8.148	15.111
polished_periodic_gru_sequence_Fw	Selected candidate	0.001195	0.001461	2.559	5.193
polished_periodic_mlp_harmonic_Fw	Selected candidate	0.001735	0.002062	3.511	7.554
polished_wave4_3_mixture_density_k3_Fw	Selected candidate	0.001528	0.001867	3.161	7.018

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
wave52b_offset_centered_shape_harmonic_Fw	Selected candidate	0.001695	0.002045	3.391	8.270

polished_rcim_model_bank_reproduction_GBM19_Fw

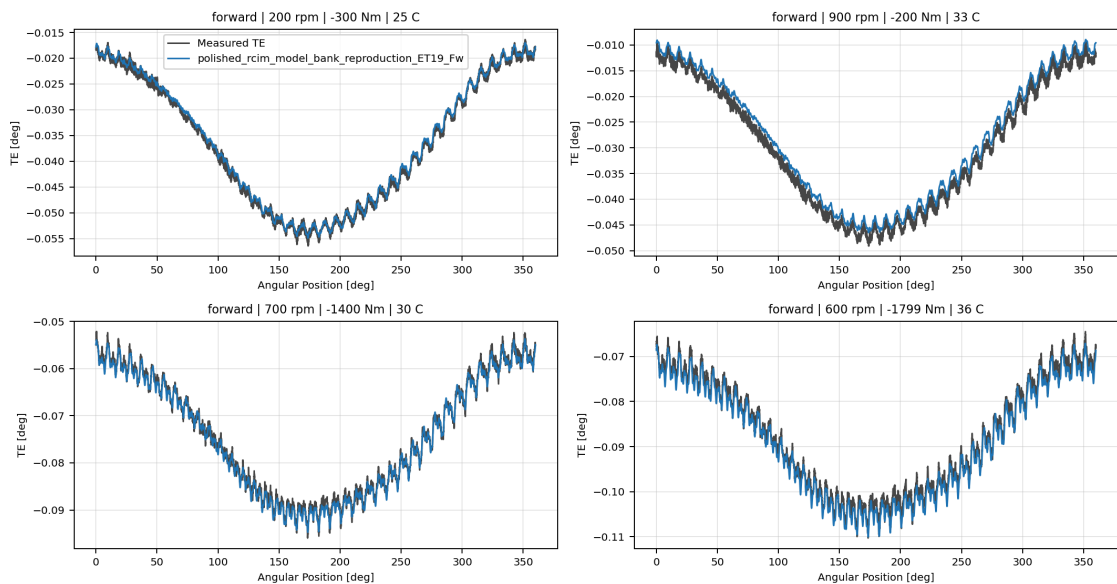
polished_rcim_model_bank_reproduction_GBM19_Fw



polished_rcim_model_bank_reproduction_GBM19_Fw measured-versus-predicted TE collage

polished_rcim_model_bank_reproduction_ET19_Fw

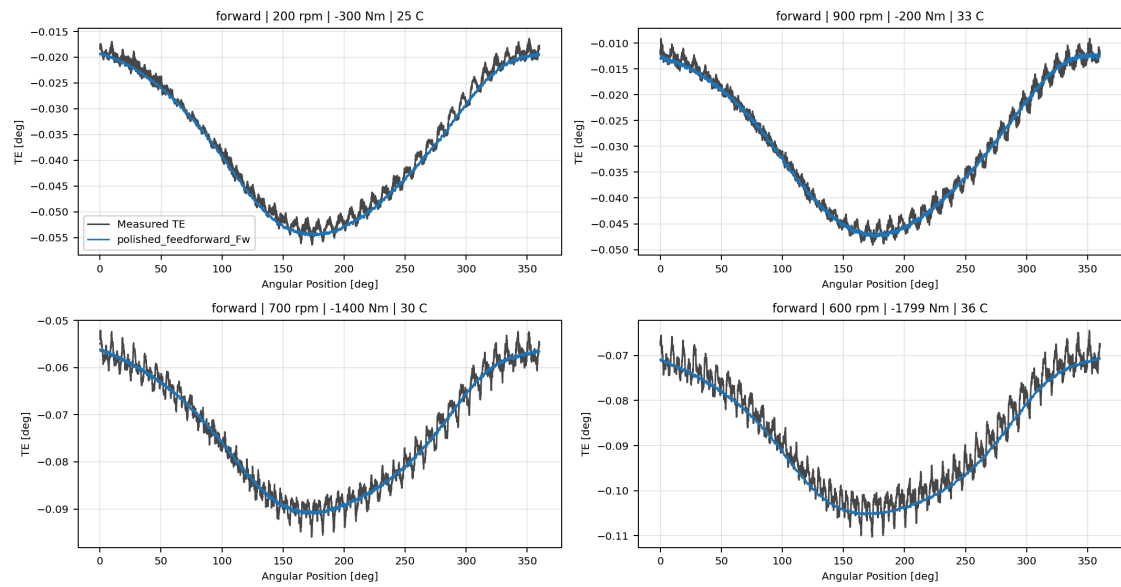
polished_rcim_model_bank_reproduction_ET19_Fw



polished_rcim_model_bank_reproduction_ET19_Fw measured-versus-predicted TE collage

polished_feedforward_Fw

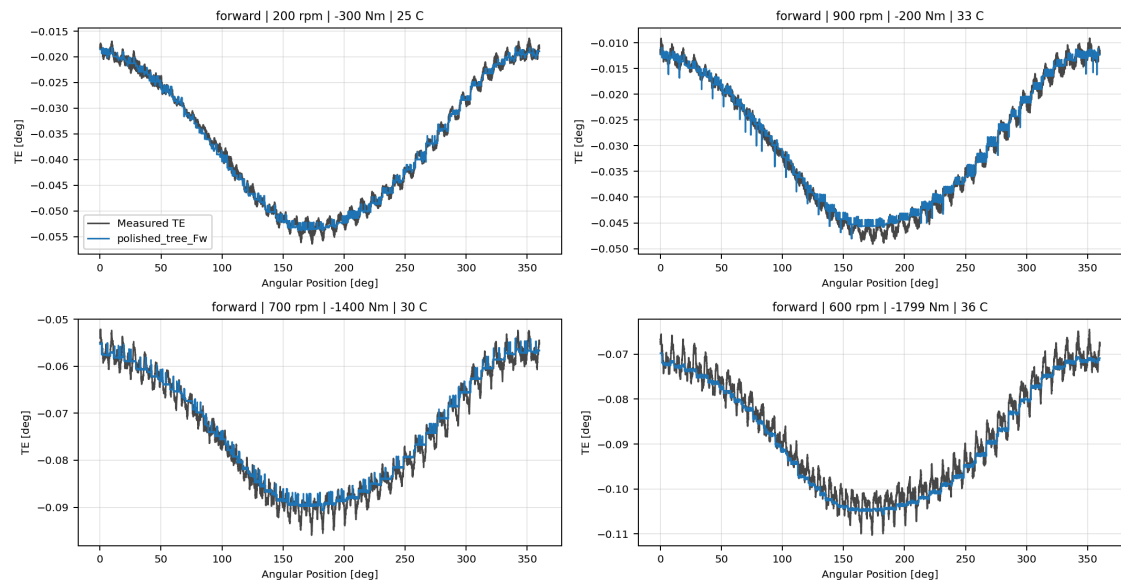
polished_feedforward_Fw



polished_feedforward_Fw measured-versus-predicted TE collage

polished_tree_Fw

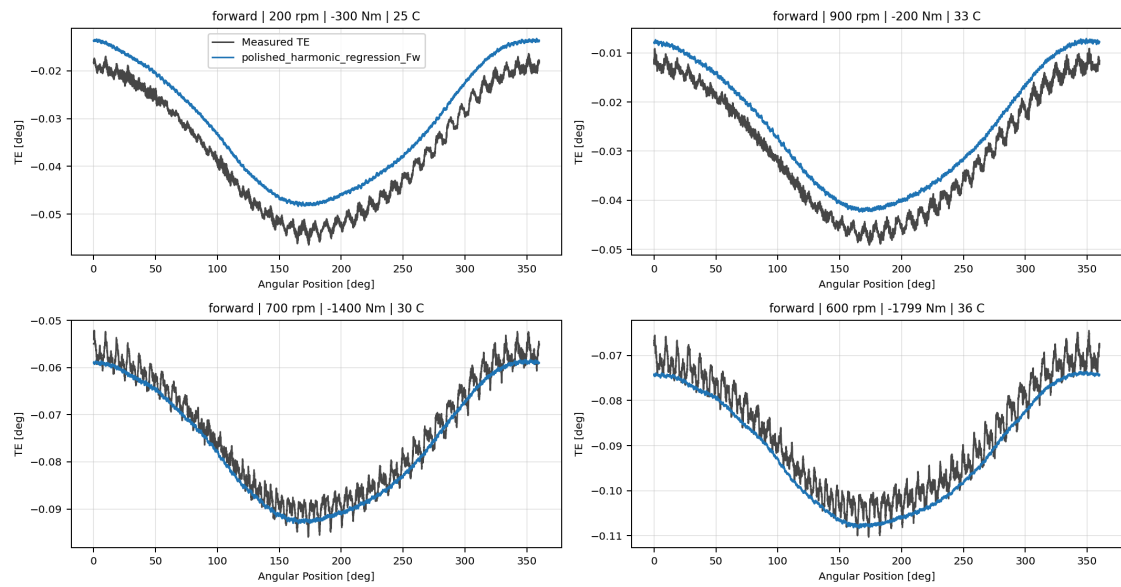
polished_tree_Fw



polished_tree_Fw measured-versus-predicted TE collage

polished_harmonic_regression_Fw

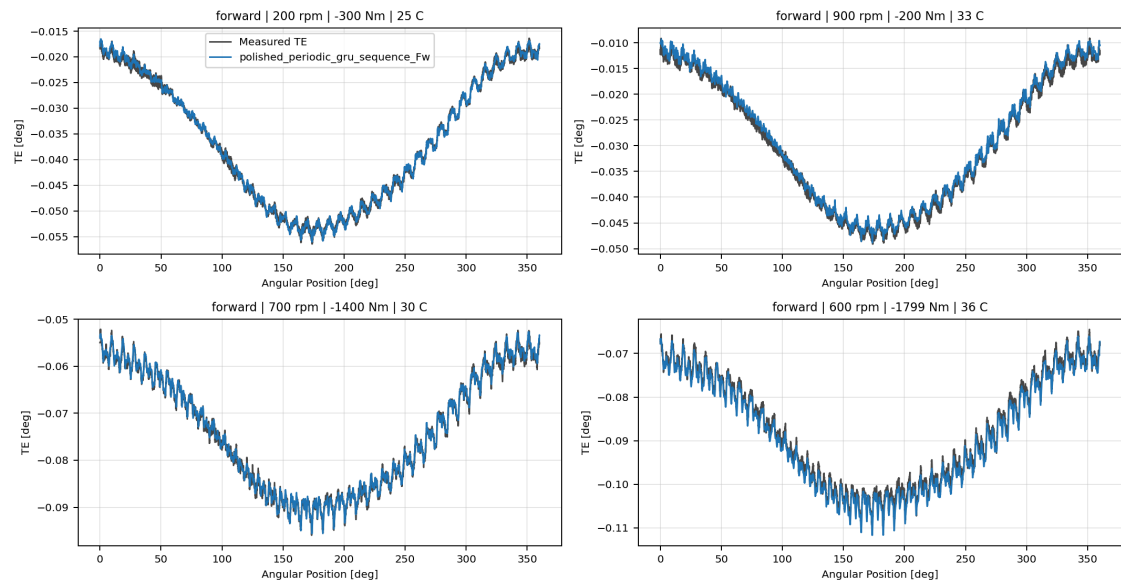
polished_harmonic_regression_Fw



polished_harmonic_regression_Fw measured-versus-predicted TE collage

polished_periodic_gru_sequence_Fw

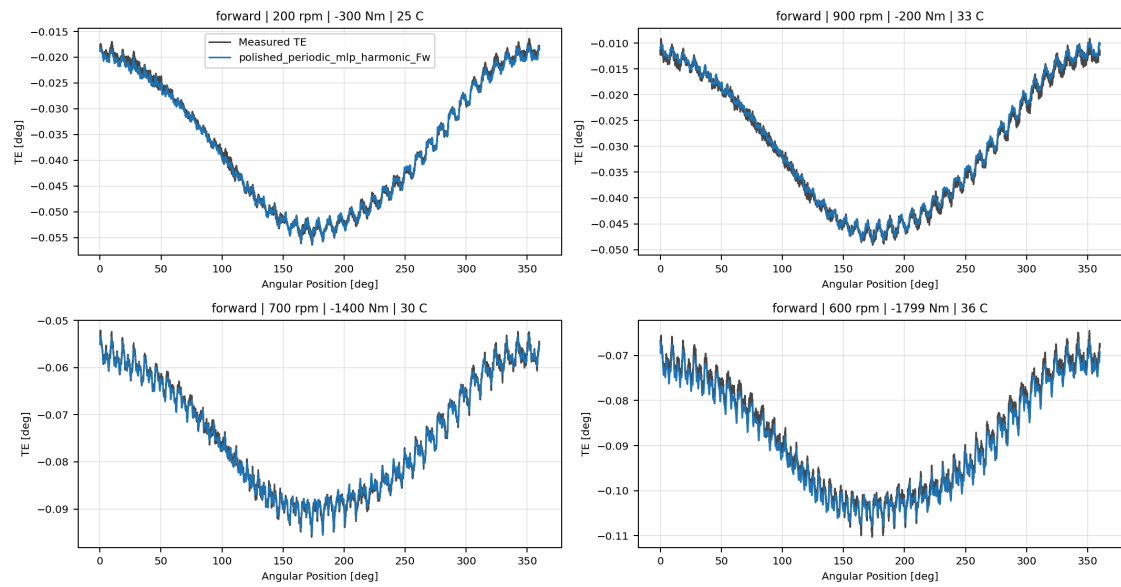
polished_periodic_gru_sequence_Fw



polished_periodic_gru_sequence_Fw measured-versus-predicted TE collage

polished_periodic_mlp_harmonic_Fw

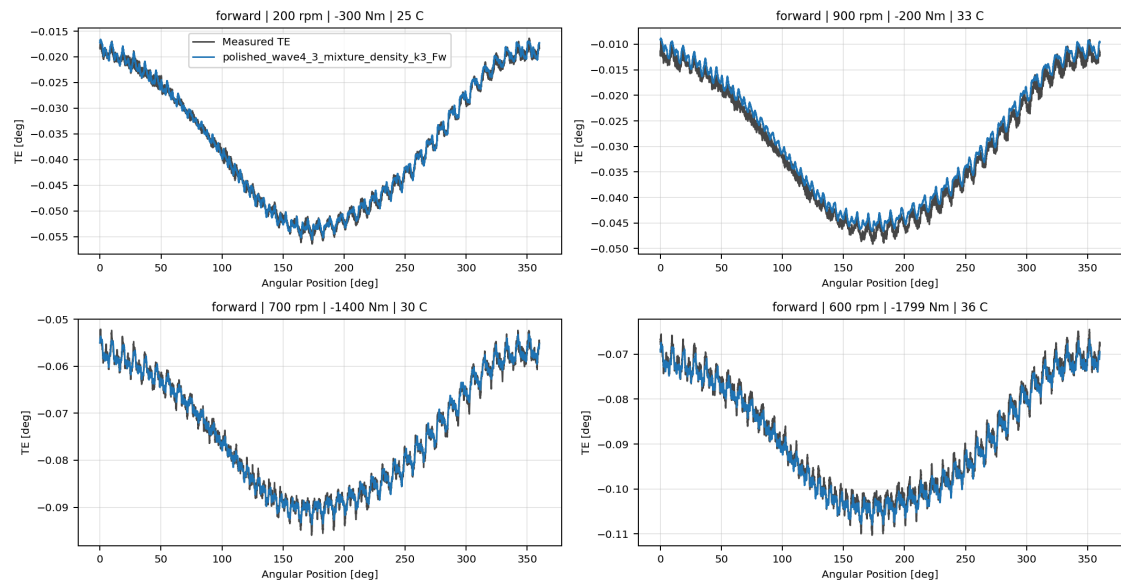
polished_periodic_mlp_harmonic_Fw



polished_periodic_mlp_harmonic_Fw measured-versus-predicted TE collage

polished_wave4_3_mixture_density_k3_Fw

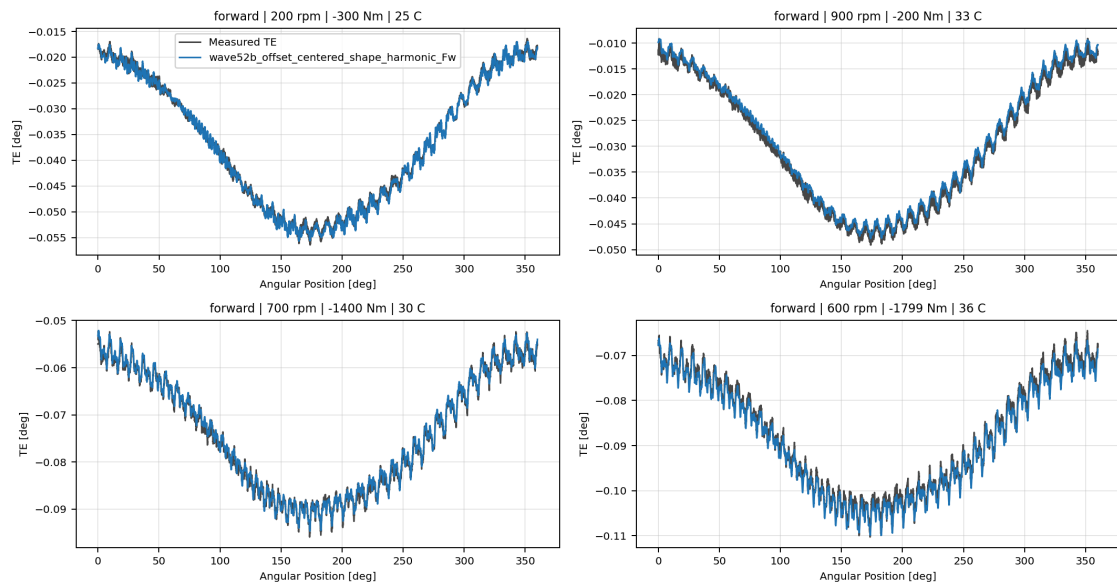
polished_wave4_3_mixture_density_k3_Fw



polished_wave4_3_mixture_density_k3_Fw measured-versus-predicted TE collage

wave52b_offset_centered_shape_harmonic_Fw

wave52b_offset_centered_shape_harmonic_Fw



wave52b_offset_centered_shape_harmonic_Fw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-07-01-16-59_track2_reduced_selected_model_matrix_track2_selected_polished_dataset_forward_2026_07_06\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-07-01-16-59_track2_reduced_selected_model_matrix_track2_selected_polished_dataset_forward_2026_07_06\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\paused_reduced_selected_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within each source group and direction. Directional paper-reference, Wave 1, and Wave 2.1 models are never evaluated on the opposite direction. Global Wave models remain valid on both directions and are therefore shown in the directional sections and again in the global breakdown. The `rcim_track1` forward reference banks use the opposite stored `h0` sign convention relative to the TE Curve Verification Pipeline reconstruction contract, so the TE Curve Verification Pipeline comparison applies the documented source-specific `h0` compatibility multiplier before curve reconstruction.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online Table 9 compensation benchmark.
- The report uses the saved Python model artifacts from `models/` ; ONNX parity checks remain a separate deployment-readiness task.